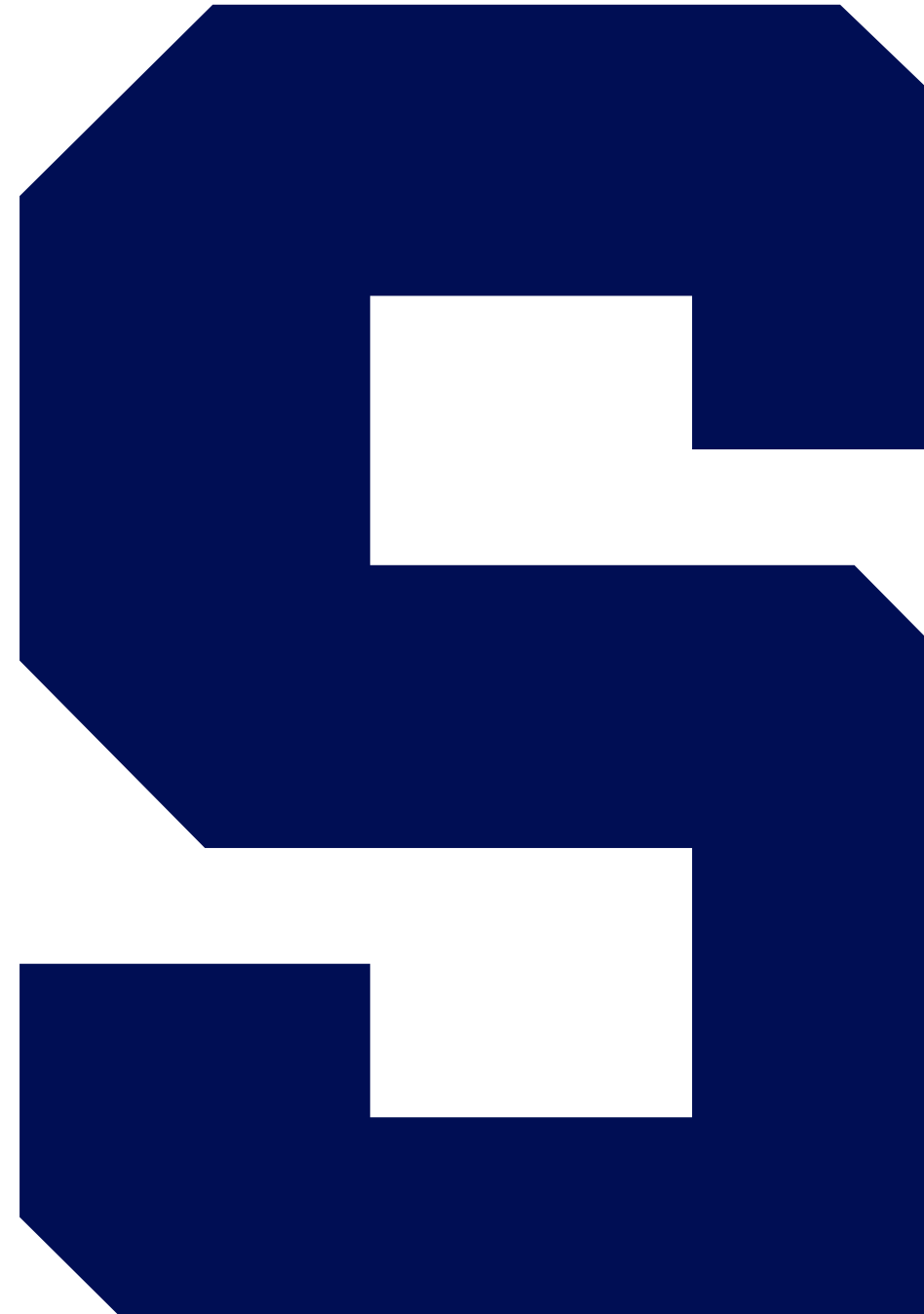
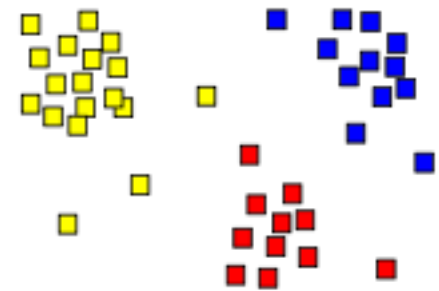


Clustering Overview



Cluster Analysis

Cluster analysis or clustering is the task of grouping a set of objects in such a way that objects in the same group are more similar to each other than to those in other groups (Wikipedia).



Clustering Applications

- Clustering is a form of **unsupervised** learning.
 - Applications:
 - Customer segmentation/recommendation systems.
 - Exploratory data analysis: learn the structure of your data.
 - Dimensionality reduction: replace feature vector x with cluster affinities.
 - Feature engineering: cluster affinities as extra features.
 - Semi-supervised learning/label propagation.
-

Different Clustering Algorithms

- There is **no single definition of a cluster**. The notion of a cluster varies with different algorithms.
- **Centroid-based models:** K-Means represents each cluster by a single mean vector.

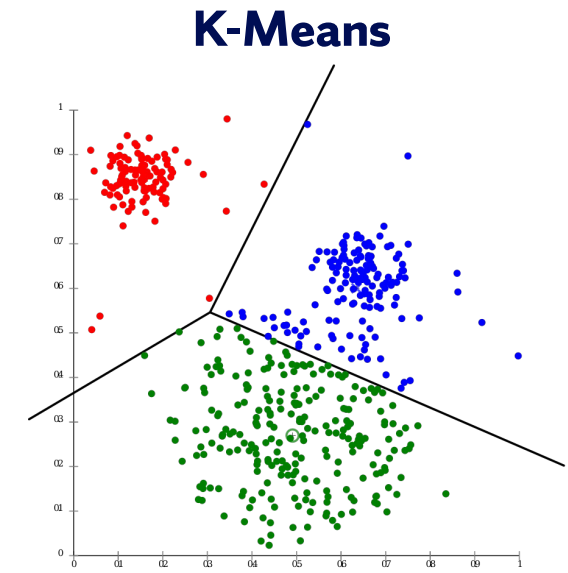


Image source: Wikipedia

Different Clustering Algorithms

- **Connectivity models:** Hierarchical clustering (e.g., HAC) builds models based on distance connectivity and builds a hierarchy of clusters.

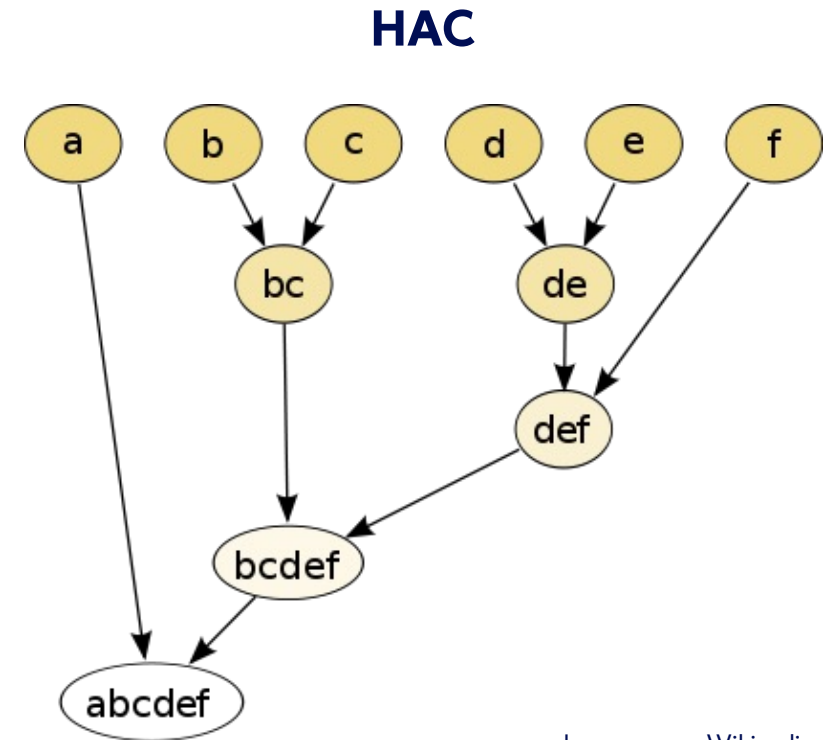


Image source: Wikipedia

Different Clustering Algorithms

- **Density models:** DBSCAN defines clusters as connected dense regions in the data space and ignores data points in sparse regions (noise or outliers).
- And other approaches.

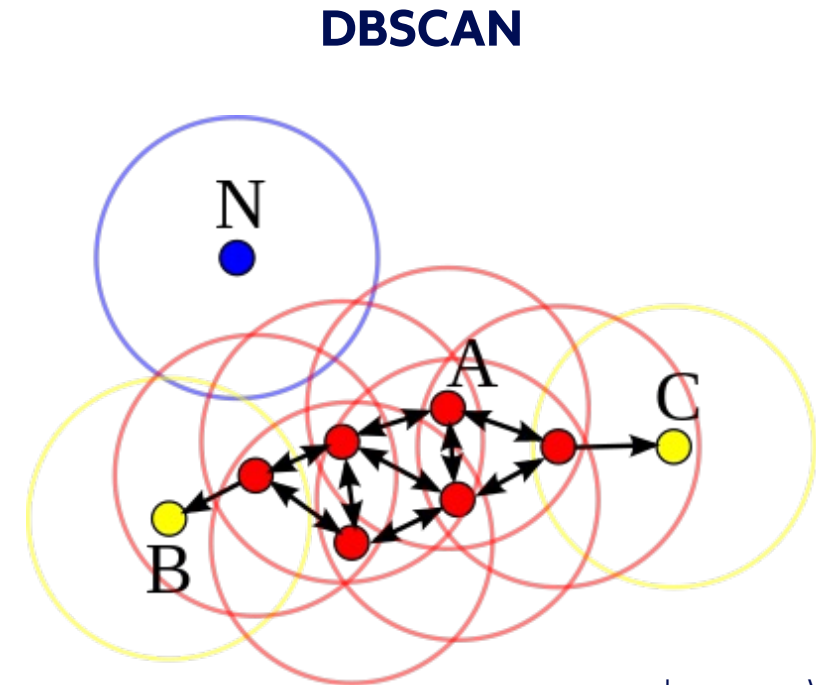
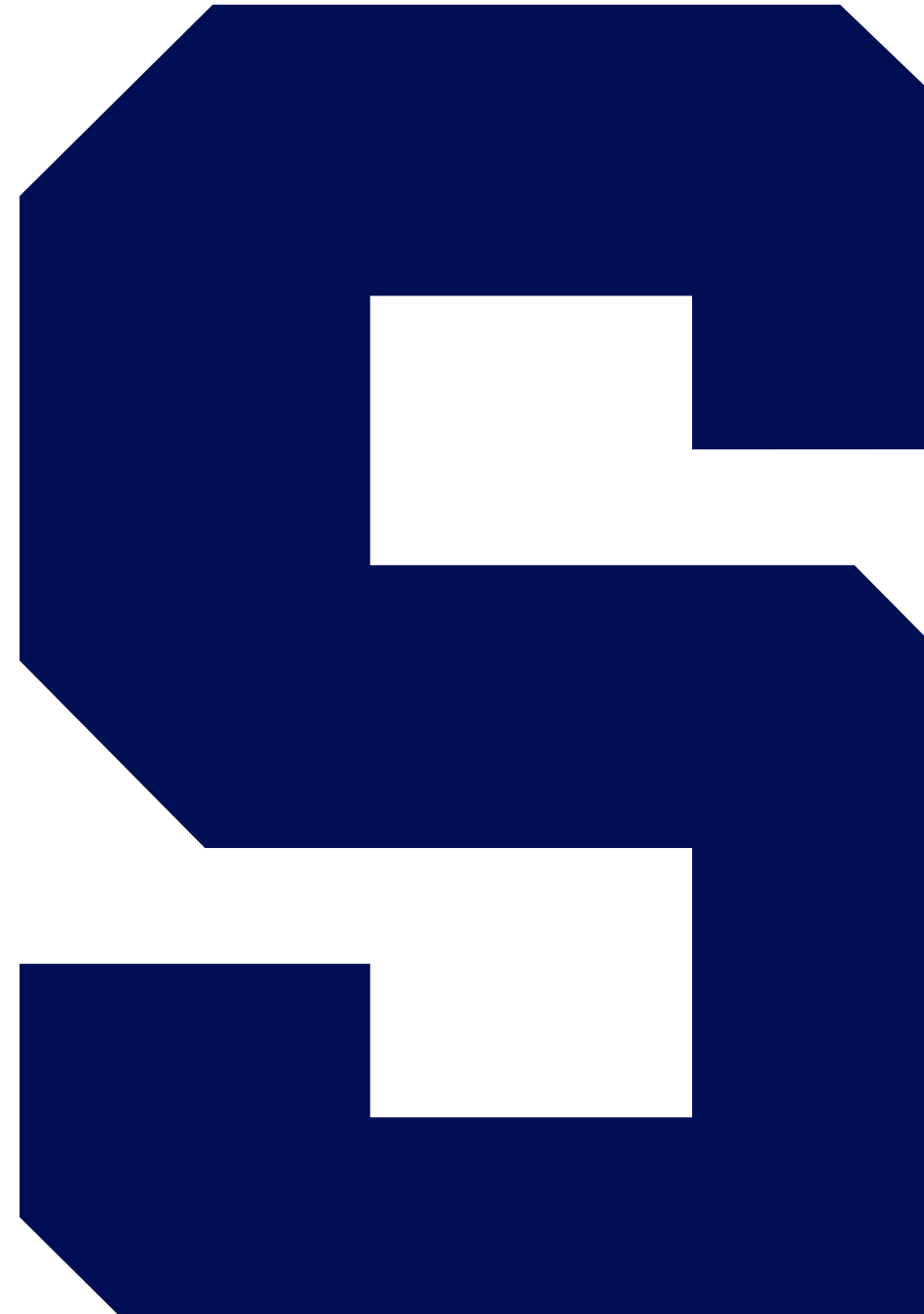


Image source: Wikipedia

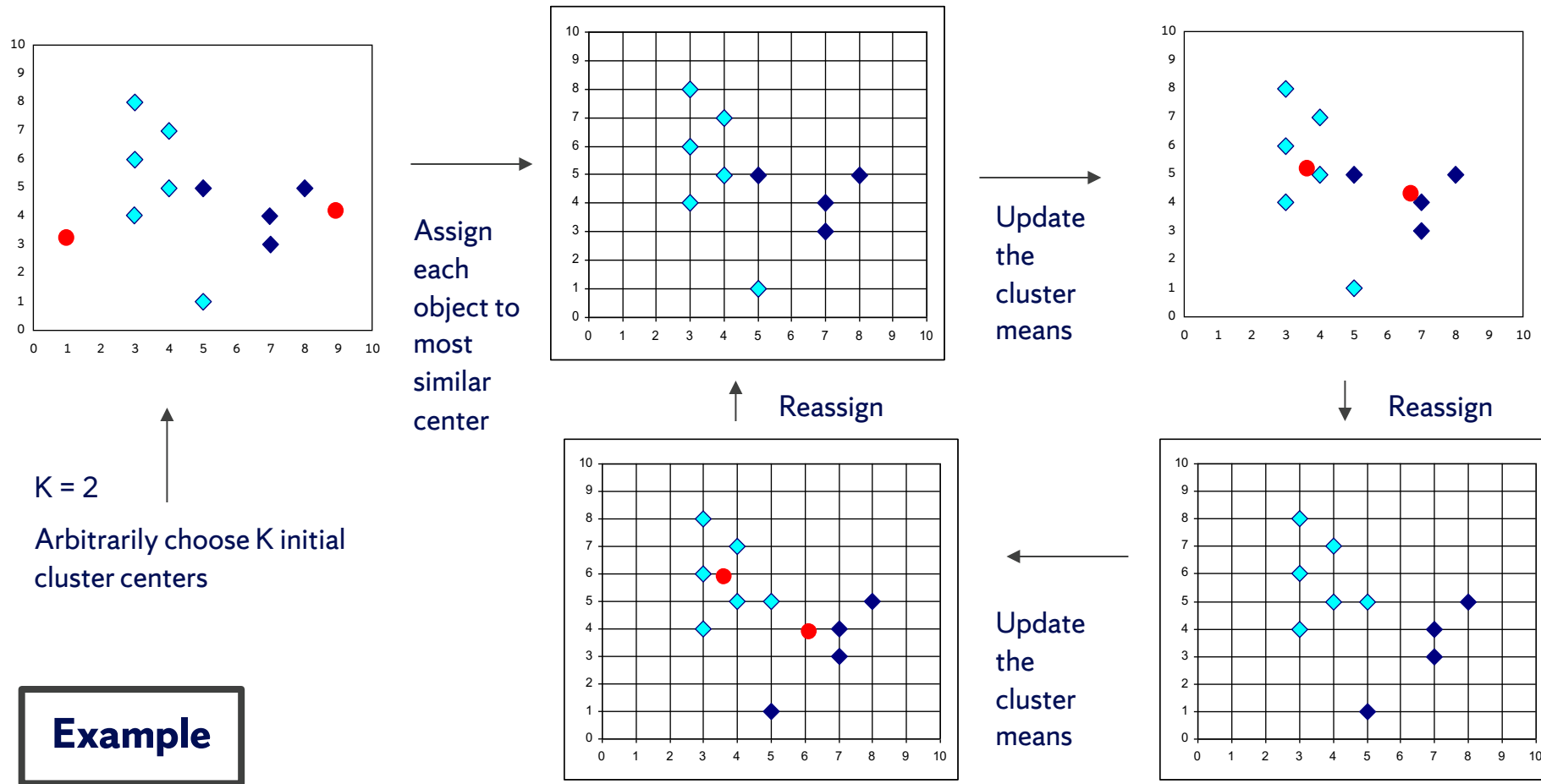
K-Means Clustering



K-Means Clustering Algorithm

- Define the “**gravity center**” of a cluster as the average of all data examples in a cluster, its **centroid**.
 - Algorithm
 - Select **K** points as the initial centroids.
 - repeat**
 - Form K clusters by assigning all points to the closest centroid, e.g., by using Euclidian distance.
 - Recompute the centroid of each cluster.
 - until** the centroids don't change
-

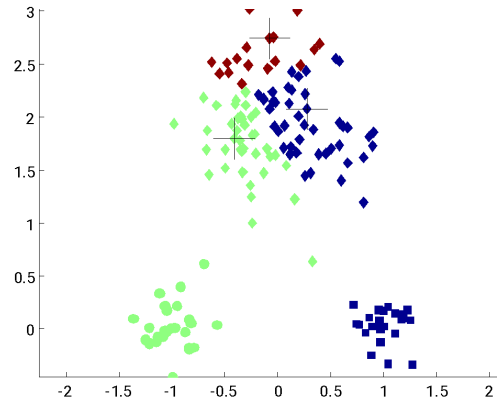
K-Means Clustering Visualized



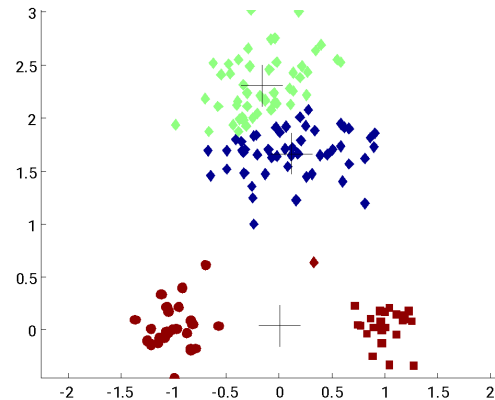
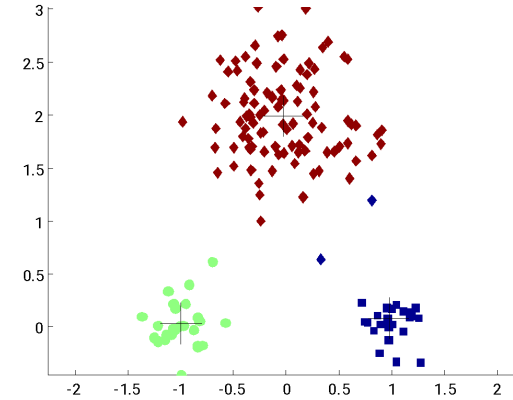
The Importance of Initial Centers

A good clustering result:

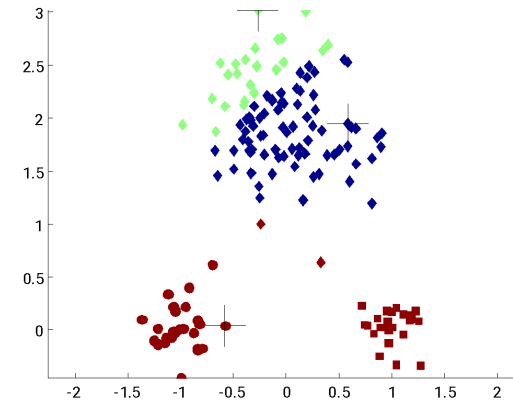
- Distance **between** clusters is **maximized**.
- Distance **within** clusters is **minimized**.



Good
➔



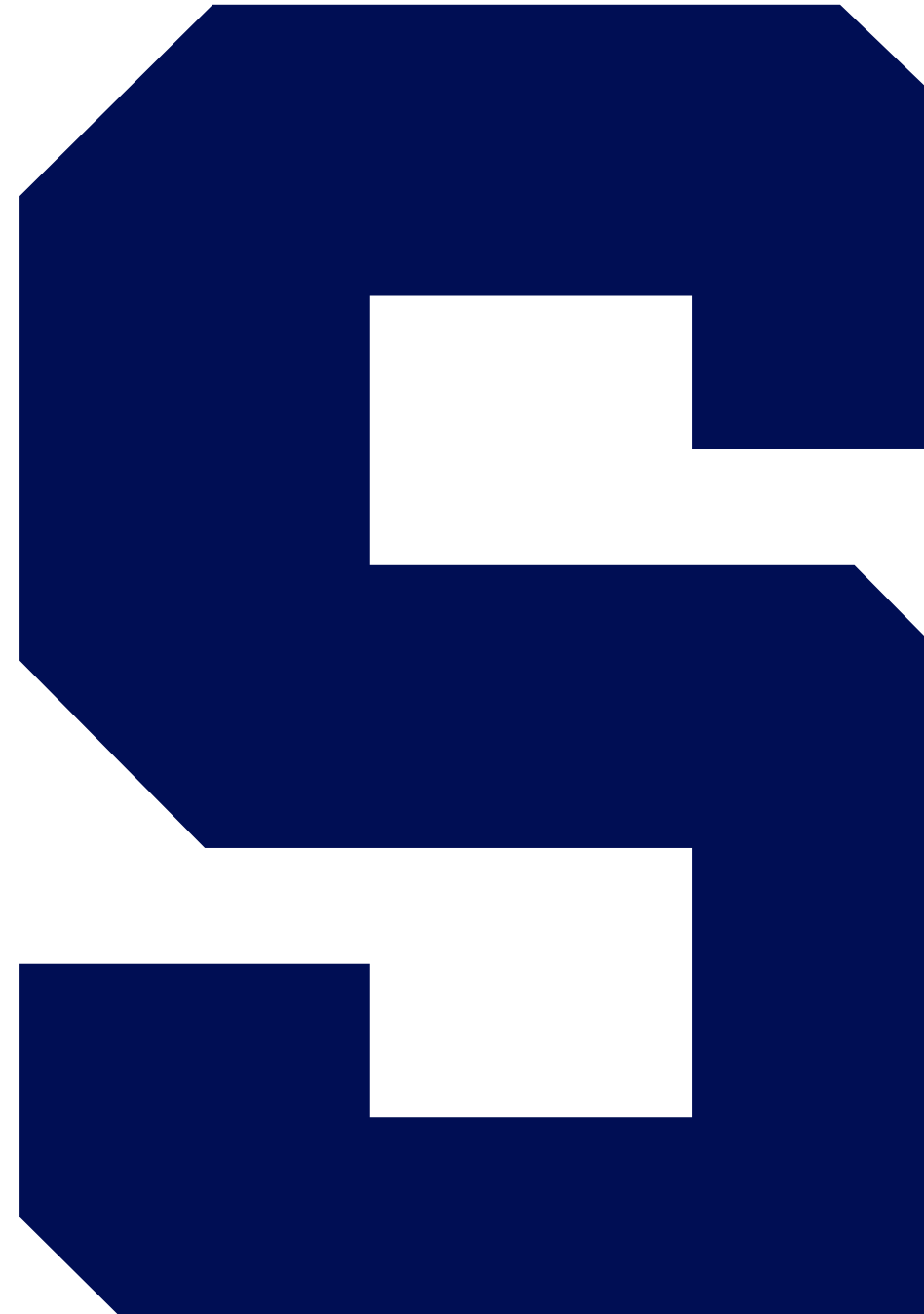
Not So
Good
➔



Methods for Choosing Initial Centroids

- **Random Initialization:** Randomly choose k data points from the dataset as the initial centroids.
 - **Forgy Method:** Similar to random initialization, but instead of picking data points as centroids, k distinct instances are chosen at random from the dataset.
 - **K-Means++:** This method aims to spread out the initial centroids as much as possible.
 - First, one centroid is chosen uniformly at random from the data points.
 - The next centroid is selected from the remaining data points with probability proportional to the square of the distance from the point to the nearest existing centroid.
 - Repeat until k centroids are chosen.
-

Cluster Evaluation



Evaluating Clustering

- Evaluating clustering is a challenging and highly **subjective** task. In most cases, there is **no "true" clustering**, and so cluster evaluation is necessarily an **exploratory** process.
 - **Domain knowledge** is very important. The "best" number of clusters according to data-driven metrics may not be meaningful for the problem you're trying to solve; oftentimes, varying the number of clusters is a diagnostic tool that will help you get a little closer to understanding your data.
 - The next most important thing to remember is the value of **visualization** — especially when clustering, visual inspection can help you understand not only the performance of clustering, but what sorts of clustering techniques are going to be useful.
 - We will explore a few common techniques for interpreting and examining clusters.
-

K-Means and Inertia

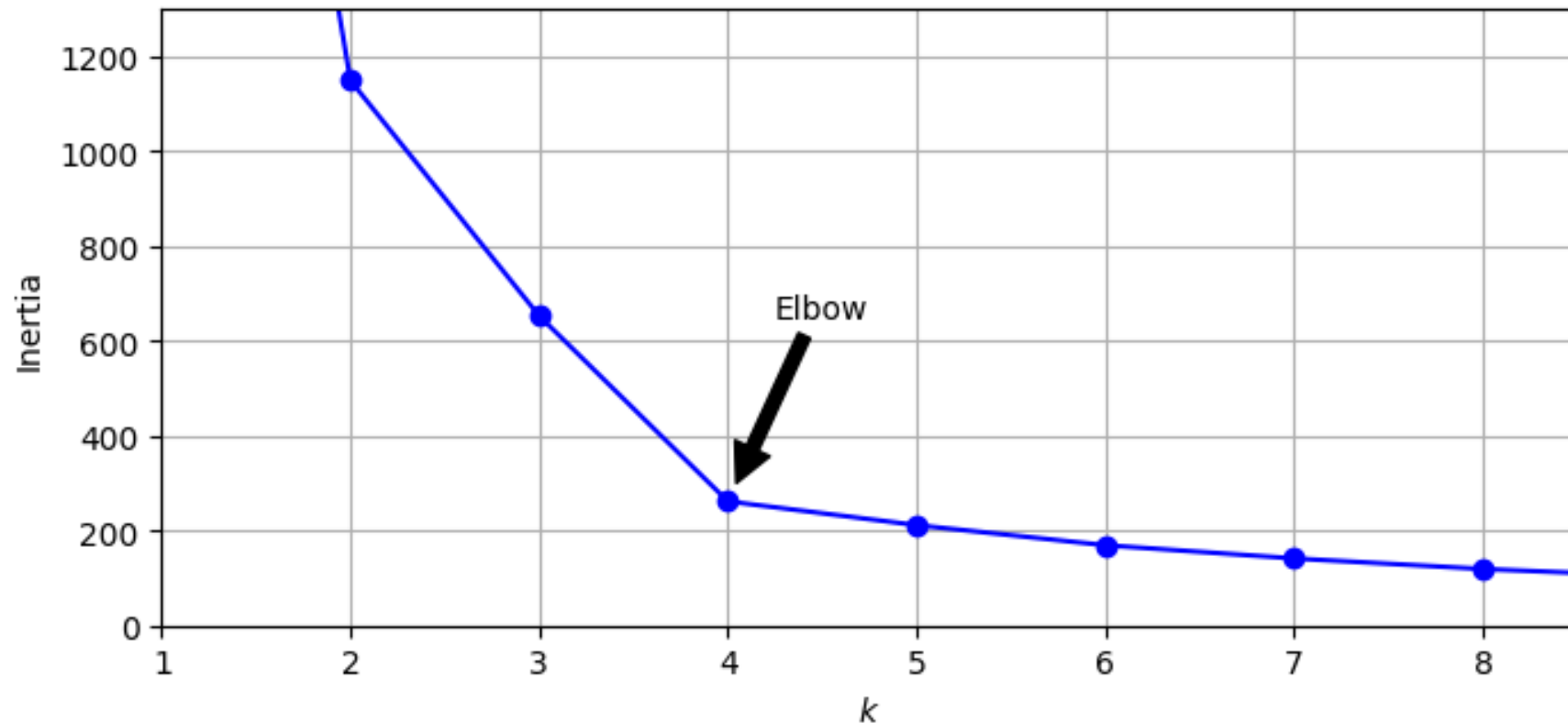
- A metric for evaluating a K-Means clustering is called **Within Cluster Sum of Squares (WCSS)**, also known as **inertia**.

$$\text{WCSS} = \sum_{i=1}^k \sum_{x \in C_i} ||x - \mu_i||^2$$

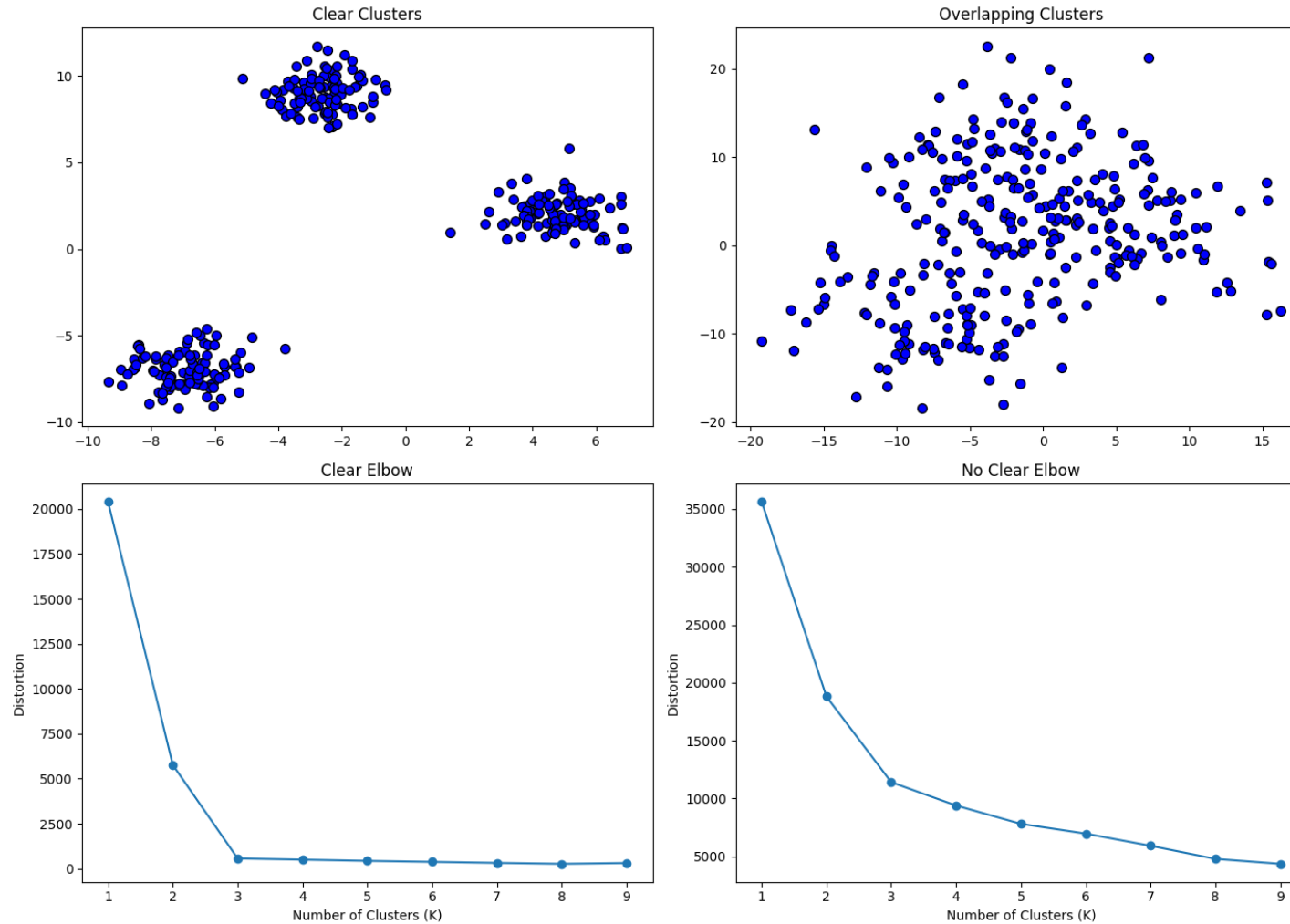
- Inertia measures **how tightly data points in a cluster are grouped together**. It's a way to quantify the compactness of clusters: lower WCSS values indicate more compact clusters, where data points are closer to the cluster center.
 - **Lower is better** but be careful: we cannot simply take the value of k that minimizes the inertia, since it keeps getting lower as we increase k .
-

K-Means and the Elbow Method

The elbow method involves plotting inertia as a function of k and analyzing the resulting curve.



The Elbow Method with Noisy Data



The Silhouette Method

The **silhouette method** evaluates how well data points fit within their assigned clusters by measuring how similar they are to their own cluster compared to other clusters.

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

$a(i)$ = Average distance from the i th point to the other points in the same cluster.

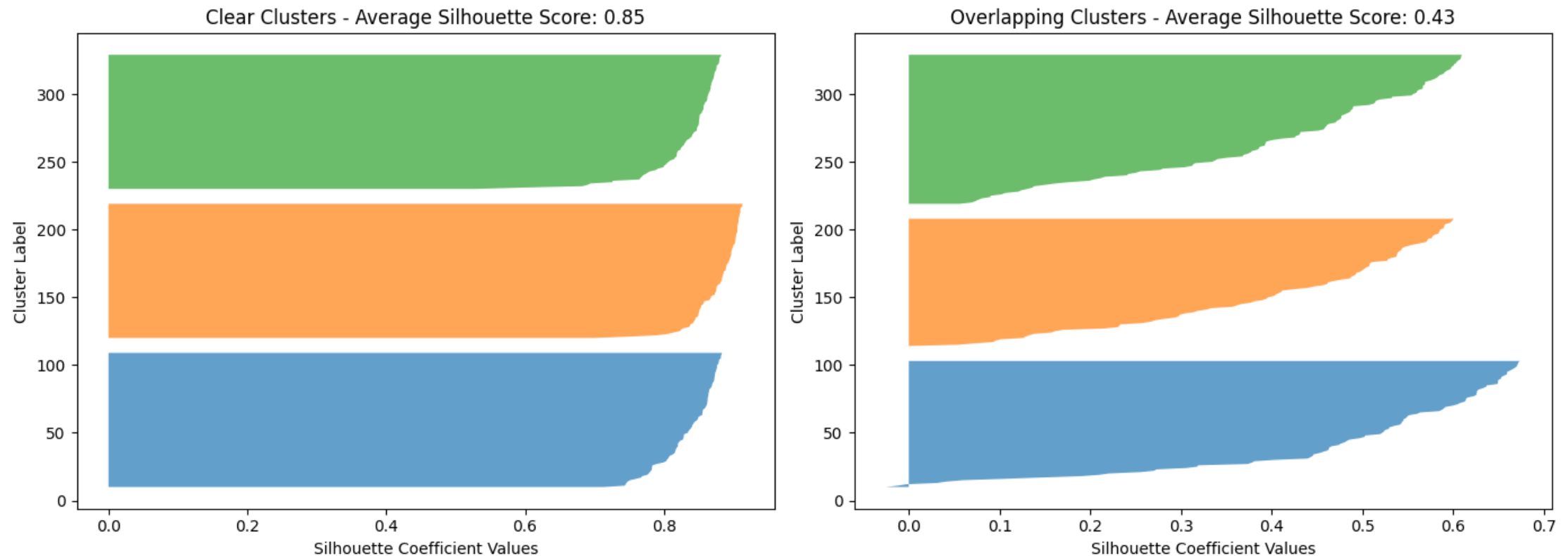
$b(i)$ = Lowest average distance from the i th point to the points in the other clusters, minimized over clusters.

Silhouette scores range from -1 for incorrect clustering to +1 for highly dense clustering.

Higher silhouette scores indicate that the data points are **well-matched** to their own cluster and poorly matched to neighboring clusters.

Visualizing the Silhouette Score

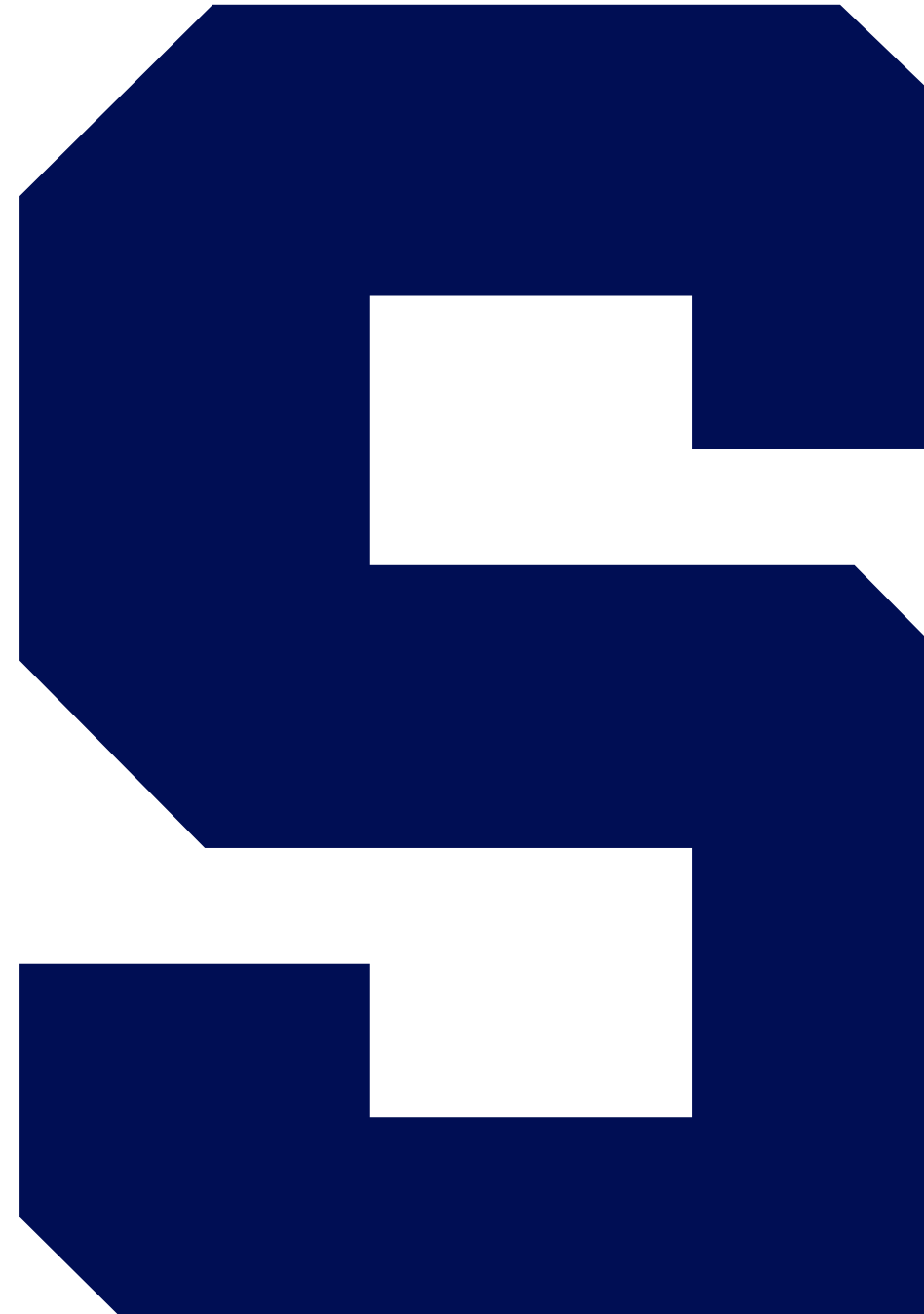
We can use the elbow method for average silhouette scores for different values of k (higher is better), or we can use a silhouette plot, where a **high average width** indicates a good clustering.



Other Techniques for Cluster Evaluation

- The **Davies-Bouldin Index (DBI)** in clustering assesses the quality of clusters by comparing the ratio of the within cluster scatter to the between cluster separation for all cluster pairs.
 - **Lower scores indicate better clustering**, where clusters are more distinct and compact.
 - **Visualize** cluster results using dimensionality reduction techniques such as PCA and t-SNE. Visualization best practices.
 1. **Standardization**: Always standardize your data before applying dimensionality reduction techniques.
 2. **Multiple Views**: Use multiple methods for dimensionality reduction to get different views of the data.
 3. **Color Code**: Use color coding to differentiate between clusters and to compare them against known labels, if available.
 4. **Pair with Metrics**: Always use visual inspection in conjunction with other evaluation metrics for a more robust assessment.
-

Advanced Clustering Techniques



Hierarchical Agglomerative Clustering (HAC)

Hierarchical Agglomerative Clustering (HAC) groups data into clusters by starting with each data point as its own cluster and repeatedly combining the closest pairs of clusters.

Advantages

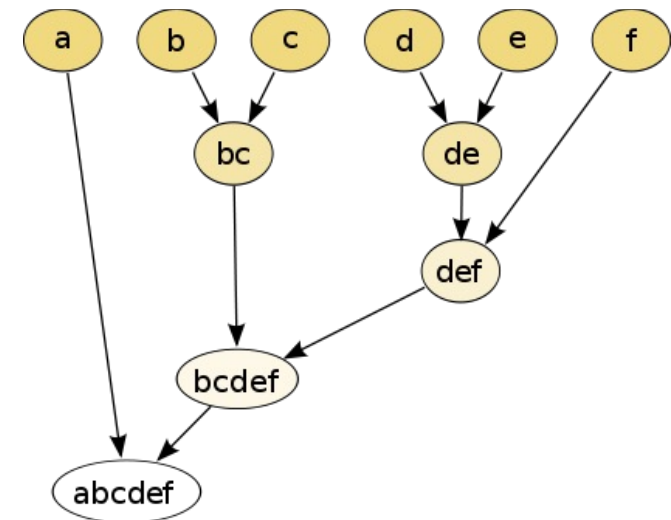
- No need to specify the number of clusters; can be determined by inspecting the dendrogram.
- Flexibility with distance metrics: Euclidean, Manhattan, cosine.
- Useful if data is inherently hierarchical.

Limitations

- Computationally expensive for large datasets.
- Irreversible: Once a decision is made to combine two clusters, it cannot be undone.

Practical Tips and Considerations

- Good practice to scale the data so that all features have the same weight.
- HAC can be sensitive to noise and outliers. Preprocessing can help.
- Choosing the right linkage can significantly impact the resulting clusters (single, complete, average)



Density-Based Clustering

DBScan (Density-Based Spatial Clustering of Applications with Noise) groups together data points that are close to each other based on a distance measurement (usually Euclidean distance) and a minimum number of points. It also marks as outliers the points that are in low-density regions.

How It Works

The algorithm starts with an arbitrary point and retrieves all points in its neighborhood (defined by a hyperparameter **epsilon**). If the point is a **core point** (min. samples are within epsilon of the point), a cluster is formed. If it's a **border point**, no points are density-reachable from it, and the algorithm visits the next point. A **noise or outlier point** is one that is neither a core or border point.

This process continues until all points are processed.

Advantages

- Can find arbitrarily shaped clusters.
- Doesn't require specifying the number of clusters.
- Can mark outliers.

Limitations

- Struggles with clusters of varying densities.
- Not well-suited for high-dimensional data.

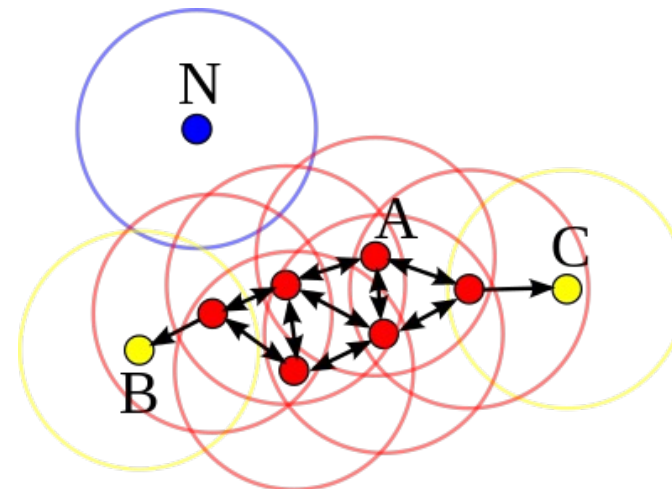
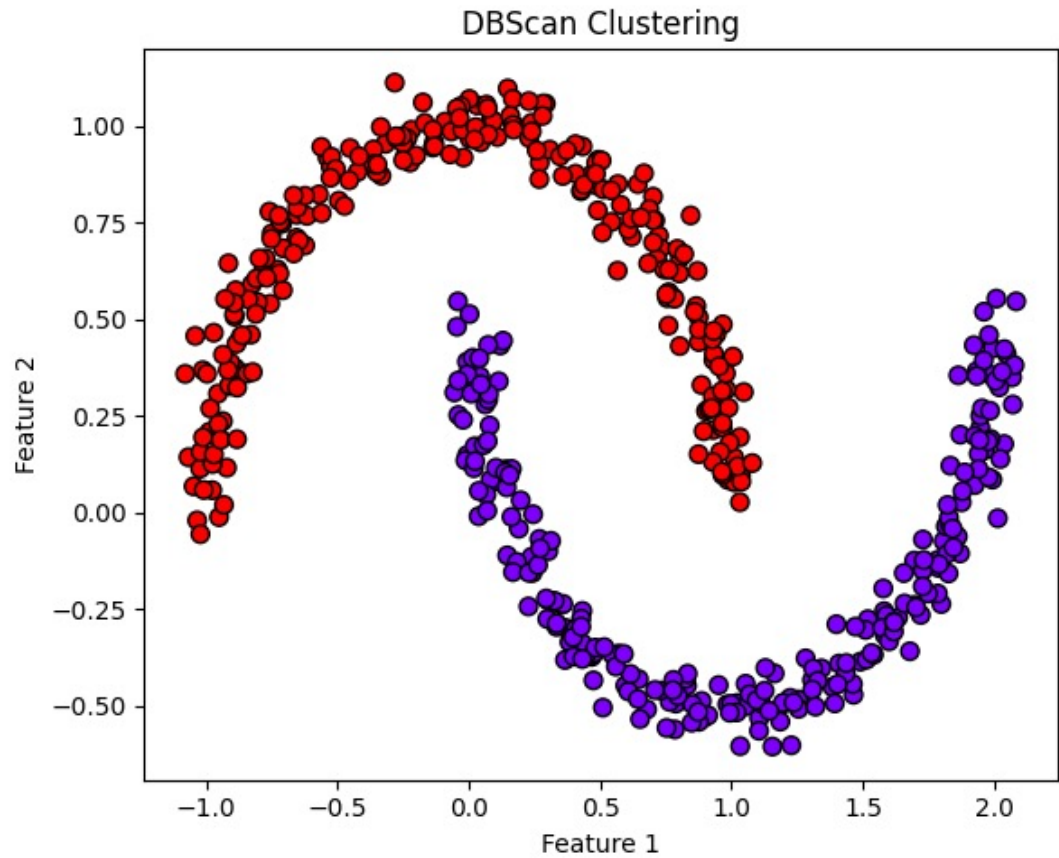
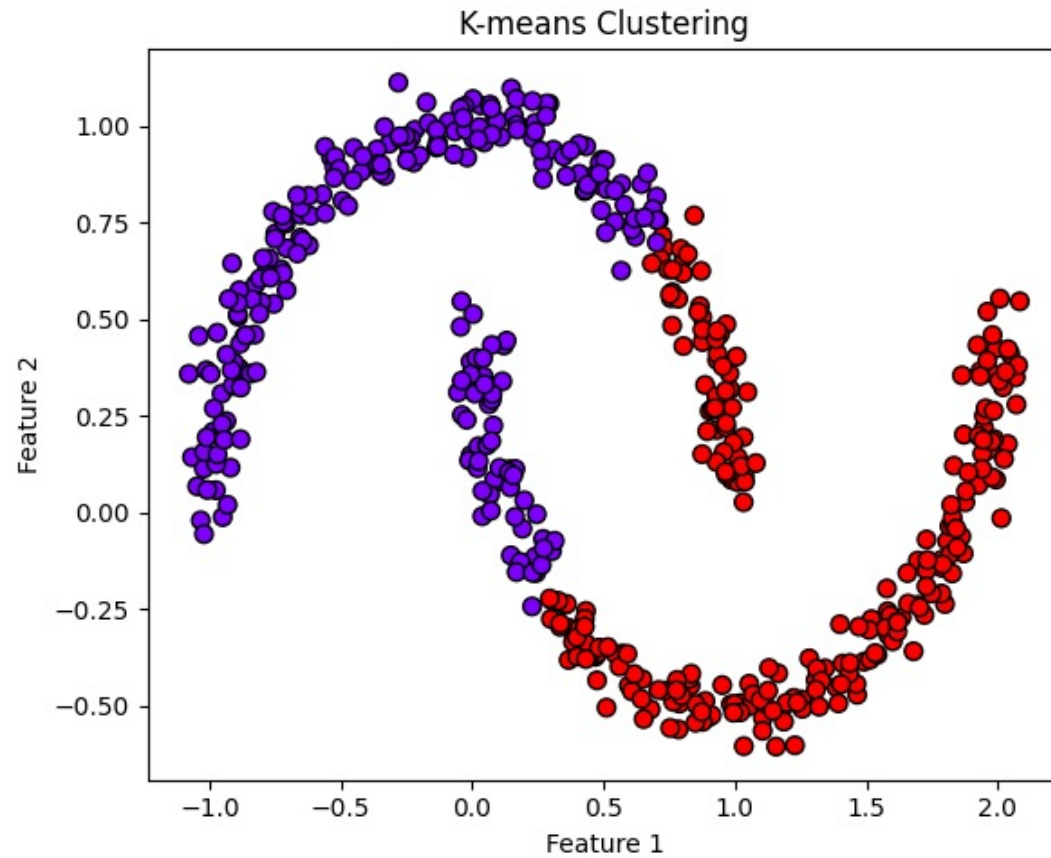


Image source: Wikipedia

Clustering the Moons Dataset



Other Advances Clustering Methods

- **Spectral Clustering:** A graph-based approach that performs dimensionality reduction before clustering in fewer dimensions.
 - **Gaussian Mixture Models:** Probabilistic models that assume data is generated from a mixture of several Gaussian distributions.
 - **Mean-shift Clustering:** A method that identifies modes (peaks) in the data distribution using kernel density estimation.
-